



An experimental analysis of the gear tooth bending strength predictions from ISO 6336 in low and high cycle fatigue

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Abstract

Gears are mission critical machine elements used for primary and accessory power transmission in many industrial, automotive and aerospace mechanical systems. In normal operation gear teeth experience torque dependent cyclic loading which can initiate and propagate fatigue cracks. One of the most prevalent manifestations of this fatigue action are fractures through the tooth root resulting in tooth separation, commonly called tooth bending fatigue. Probably the most cited and applied methodology for designing gears to resist these fractures is the ISO 6336 “Calculation of Load Carrying Capacity of Spur and Helical Gears” which provides analytical equations and tabulated reference data to estimate design safety factors against bending fatigue. Method A of the ISO standard considers experimental evaluations of fatigue resistance while Method B relies on tabulated reference data. It is important that gear designers understand potential differences in the resulting fatigue strength and location of failure on the tooth root (critical location) between different design methods to ensure designs are safe from fracture. However, there is little information available in English and with tabulated data that gives a comparison on the accuracy (critical location and fatigue life) between experimental methods and analytical results. Even less information is available in terms of failure due to cyclic loads exceeding the yield strength (Low Cycle Fatigue), even if such type of loading should be carefully analyzed in case of applications involving severe loading, such as aerospace applications, or to account for unexpected load peaks. This work explores bending fatigue failure in gears under cyclic plastic loading. A gear Single Tooth Bending (STB) fatigue methodology is used to perform Low Cycle Fatigue (<10,000 cycles) tests to complement existing High Cycle Fatigue STB data on case-carburized gear specimens made from AISI 8620 (20NiCrMo2-2). The critical locations and the fatigue strengths are compared between the experimental methodology and the analytical results of ISO 6336 Method B, providing insight about the accuracy of ISO 6336 when cyclic plasticity is involved.

1 Introduction

When sizing gears, a variety of failure modes is to be considered and prevented to achieve a safe and reliable design. Most of them occur due to the nature of gear operation, which is based on the transmission of power through two meshing profiles [1]. This involves torque dependent cyclic loading and contacts of gear teeth, which causes their gradual degradation until final failure. The two most prevalent manifestations of such fatigue action are fracture

driven by the contact pressure on the involute portion of the tooth, commonly called pitting, and fracture through the tooth root resulting in tooth separation, commonly called Tooth Bending Fatigue (TBF) [2]. Indeed, tooth roots represent the tooth region that is mostly affected by bending stresses that are generated by the transmission of loads. Such stresses, being cyclically repeated, can cause the nucleation of cracks, which, cycle after cycle, propagate inside the tooth root, until the resistant section is too small and sudden failure occurs [2]. Such abrupt failure mode makes TBF particularly dangerous, as it could lead to unexpected and potentially catastrophic outcomes, both from the economic and from the safety points of view.

For this reason, specific experimental procedures and standards prescribing design rules were developed [3, 4]. Probably the most cited and applied methodology for designing gears to resist these fractures is the ISO 6336 “Calculation of Load Carrying Capacity of Spur and Helical

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Gears” [3], which provides analytical equations and tabulated reference data to estimate the design safety factor against both contact and bending fatigue mechanisms. Method A of the ISO standard considers experimental evaluations of the fatigue resistance while Method B relies on the tabulated reference data. Experimental evaluations of TBF are carried out by testing gears either in the meshing gears configuration or through pulsator testing, also referred to as Single Tooth Bending (STB) fatigue testing [5]. While tests conducted in the meshing gears configuration are more representative of the real way gears are employed, they are time and resource consuming. This is mainly because in such tests all teeth are loaded and failure of a single tooth terminates the test for the entire gear test specimen. Conversely, in STB tests multiple teeth belonging to the same gear can be failed, since they are loaded individually [6]. Indeed, in STB experiments, gears are mounted on dedicated fixtures and placed between two anvils that apply a pulsating load, compromising only two teeth per test and allowing for multiple test per gear specimen. Details about STB testing can be found in SAE recommended practice for STB fatigue [7].

The adoption of different experimental testing methods and of different designs guidelines strongly influences the distribution of stresses in tooth roots, hence influencing STB fatigue behavior of gears and producing different outcomes in terms of fatigue strength and critical location (i.e., location of origin of the crack producing the final failure) [8, 9]. It is important that gear designers understand potential differences in the resulting critical location and fatigue strength between the different design methods to ensure designs are safe from fracture. However, there is little information available in English and with tabulated data that gives a comparison on the accuracy (critical location and fatigue life) between experimental methods and the analytical results. Even less information is available as far as failures due to cyclic loads that exceed the yield strength causing plasticity, i.e., Low Cycle Fatigue (LCF) are concerned. Indeed, while most applications where gears are employed involve the loading of teeth only in the elastic field, i.e., High Cycle Fatigue (HCF) of gear teeth, in some cases loads exceed the yield strength of materials and plastic cyclic loading occurs, i.e., LCF of gear teeth. This mostly occurs in industrial fields such as the aerospace sector, where gears experience loading levels high enough to significantly deplete their lifespan within a relatively small number of cycles [10, 11]. However, unexpected overloads might occur in many other applications. In all these cases, standards for gear design such as ANSI/AGMA 2001-D04 state: “*The use of this standard at bending stress levels above those permissible for 10^4 cycles requires careful analysis*” [4]. ISO 6336:2019, in the section devoted to the calculation of tooth bending strength (Sect. 3), reports: “*This part of ISO 6336 does not apply*

at stress levels above those permissible for 10^3 cycles, since stresses in this range may exceed the elastic limit of the gear tooth” [3].

The result is that data about the LCF behaviour of gear teeth is sparse and left to the testing campaigns of individual scholars and research institutions. The first results in these terms can be found in the technical report of McIntire et al. [12], where sixteen different gear geometries were tested at various loads, including examples exceeding the yield strength of the material and inducing failure in less than 10^4 cycles. Gears were made from AISI 9310 steel, were case-carburized and were tested in STB conditions. Moore [13] obtained S–N diagrams in the LCF region, by testing spur gears (made from AISI 4340M and AISI 9310) and bevel gears (made from AISI H-11). In the technical report of Heath et al. [14], gears made from X53, CBS 600, M50NIL, Maraging 300, and AISI 9310 steel were tested in STB symmetric conditions. However, tests performed with loads inducing cyclic plastic loading were only a minor portion of the study. Handschuh et al. [11] performed LCF tests on AISI 9310 spur gears, through pulsator testing under symmetric loading and with load ratio $R=0.1$. They measured cycles to crack initiation and final failure and performed a preliminary Finite Elements Method (FEM) analysis. Ollson et al. [15] performed both HCF and LCF tests on two spur gears of different sizes, made from Ovako 253C steel, observing a small size effect on the fatigue life. This brief survey of the literature about STB tests on gears in LCF regime proves the scarce availability of data of such kind, despite the relevance of applications where such phenomena take place. Sources reporting LCF STB data are little and they focus on different materials.

In light of this, the current work explores bending fatigue failure in gears under cyclic plastic loading. The aims are primarily two. The first one is to enrich the current literature with additional LCF STB data of gears. This is done by performing LCF STB tests to complement existing HCF STB data on case-carburized gear specimens made from AISI 8620. Such steel represents a material that is commonly used in manufacturing case-hardened gears. At the same time, almost no data is available in the literature about its LCF STB behavior. The second goal is that of comparing the critical location of failures (i.e., the initial point along the tooth root where cracks nucleate and then propagate from there) as well as the fatigue strengths obtained experimentally versus those estimated by ISO 6336 Method B, providing insight about the accuracy of ISO 6336 when cyclic plastic loading is involved.

2 Test specimens and methodology

In this work, eighteen teeth of case-carburized spur gears made from AISI 8620 (20NiCrMo2-2) were tested in STB fatigue conditions, in order to define their behavior under cyclic plastic loading and measure the critical location where cracks originated in tooth roots. AISI 8620 is a low-carbon, case-hardening steel widely used in gear manufacturing, thanks to its excellent toughness and machinability, which contribute to its widespread use in industrial applications such as wind turbines, automotive gearboxes, and heavy machinery [16, 17]. Case hardening is arguably the most common surface treatment for AISI 8620 gears. This process enhances surface hardness while maintaining a tough core, improving wear resistance, fatigue strength, and dimensional stability [18].

2.1 Gears geometry, material and heat treatment

Fundamental geometric parameters of the gears are reported in Table 1, while the gear geometry is represented in Fig. 1, compliant to SAE J1619 standard [7]. The chemical composition of the AISI 8620 steel employed to manufacture the tested gears is reported in Table 2. Gear specimens were hobbled, carburized and then ground. Both flanks and roots were ground to equivalent surface finishes. The case-carburizing treatment (Fig. 2) consisted of a carburization step (4h at 927 °C) a hardening step (2h at 816 °C), oil quenching and a freezing step (4.5h at -96 °C) and eventually, a tempering process (177 °C for 3h). The treatment produced a surface hardness of 61–62 HRC and a core hardness of 30–31 HRC. Gear accuracy was checked for helix slope and profile slope and conformed to ISO 1328 classes 1–3 for all test specimens. A tooth from one specimen was removed, embedded in resin, polished and etched in Nital solution. Figure 3 shows the resulting microstructure where clear martensite formation is evident. A case depth

Table 1 Tested gear geometry parameters.

Parameter	Symbol	Unit	Value
Normal modulus	m_n	mm	4.23
Number of teeth	z	—	34
Pressure angle	α	°	20
Face width	b	mm	25.4
Pitch diameter	D_p	mm	143.934
Root diameter	D_r	mm	131.775
Base diameter	D_b	mm	138.252

Table 2 Chemical composition of AISI 8620 (20NiCrMo2-2).

Element	C	Mn	S	Si	Cr	Ni	Mo	Cu	Al	V
Wt. (%)	0.22	0.84	0.012	0.33	0.54	0.42	0.15	0.18	0.028	0.002

Fig. 1 AISI 8620 case-carburized spur gear geometry. Imperial system units

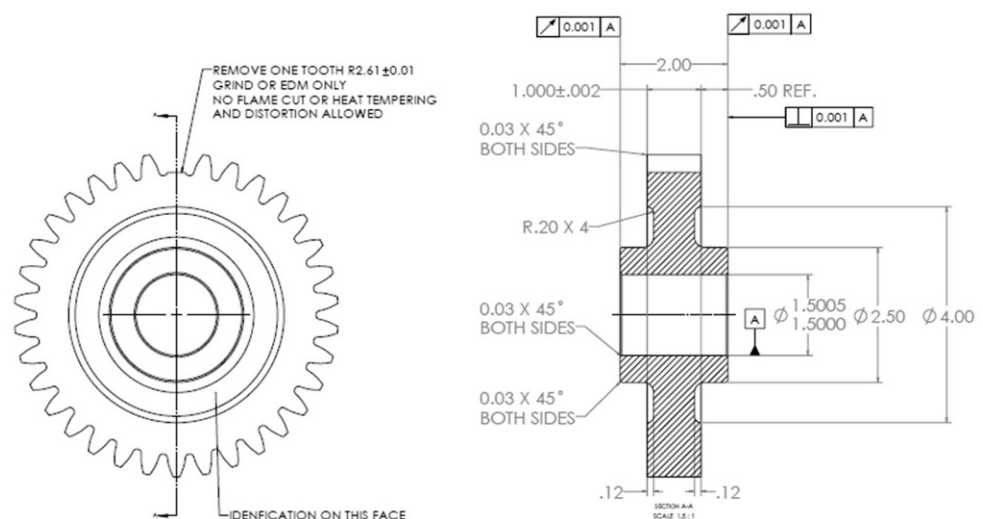
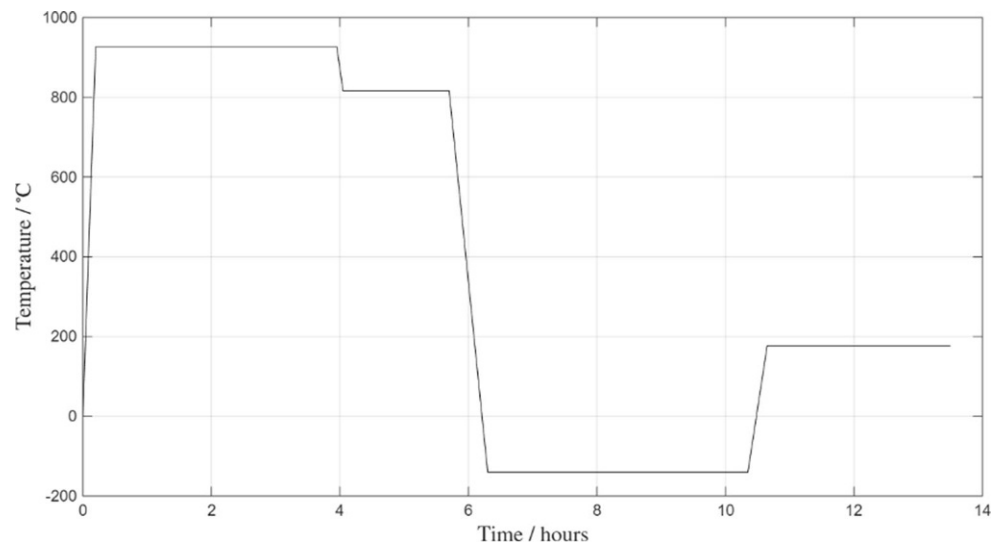


Fig. 2 Heat treatment process diagram



of 0.97 mm at 513 HV was obtained. Further microstructure identification was not pursued.

2.2 Gear fixture for STB testing and unloaded contact analysis

A STB fatigue testing framework conforming to SAE J1619 standard [7] was employed to apply a sinusoidal pulsating loading capable of producing cyclic plastic deformation. The same framework was employed in previous works of the authors [19–22]. The gear was mounted on the fixture represented in Fig. 4a. On such fixture, the gear could not rotate and had two teeth in contact, one (the test tooth) with the upper anvil (which was used to transmit the load) and one (the reaction tooth) with the lower anvil (which was used as support). The arrangement is asymmetric in which the radius of contact for the test tooth is larger than for the reaction tooth generating higher bending stresses in the test tooth as compared to the reaction tooth. This ensures that the failure will occur in the test tooth. The fixture

was mounted within a force-controlled servo-hydraulic oscillator capable of loads up to 130 kN, located at The Ohio State University. Cyclic forces were applied with a load ratio of $R=0.1$ and a frequency of 5 Hz, characterizing the STB LCF behavior of a total of eighteen teeth. In addition to the STB tests, two monotonic tests were performed to determine the Ultimate Tensile Strength (UTS) of the gear teeth. One of the two teeth employed for monotonic testing was also strain-gaged at the tooth root, in order to obtain a stress-strain diagram of the gear teeth.

To facilitate a stress analysis of the entire gear, the location of the contact points of the fixture anvils on the gear must be known. A 2D unloaded (rigid body, pure geometric) contact analysis was performed to determine the location of contact between the single tooth bending fixture and the test and reaction teeth. A cross section in the transverse plane was considered and the rotational constraint of the upper portion of the fixture considered along with the lower geometry of the fixture supporting the reaction tooth. The rigid body approach of the reaction tooth to the fixture lower anvil was solved to determine the angular position of the gear within the fixture then the final rigid body approach of the fixture upper angle to the test tooth solved for to determine the contact point. The contact point solution is plotted in CAD shown in Fig. 4b. These contact points will be utilized in more conventional gear analysis routines to determine tooth root stresses, which are explained in the next section.

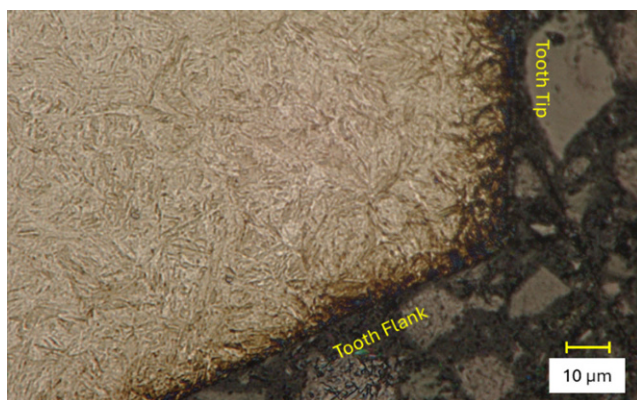
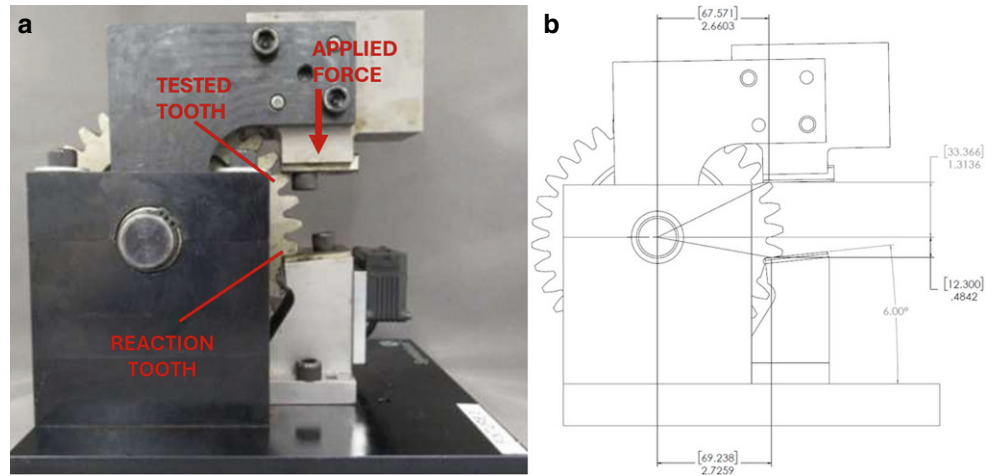


Fig. 3 Micrograph of carburized gear tooth bending fatigue specimen

2.3 Loaded tooth contact analysis and bending stress analysis

Stress analysis for the TBF experiments can be performed according to both the ISO 6336-3 standard calculation Method B as well as using dedicated software such as Win-

Fig. 4 STB fatigue test fixture (a) and unloaded contact points (b)



dows LDP (Method A). As far as the ISO 6336-3 standard is concerned, the nominal stress at the tooth root (σ_{F0}) is calculated as (Eq. 1):

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} \cdot Y_F \cdot Y_S \cdot Y_\beta \cdot Y_B \cdot Y_{DT} \quad (1)$$

where F_t is the nominal tangential load. With regard to the other parameters,

- Y_F is the form factor, which represents the effect of gear tooth geometry on nominal root bending stress σ_{F0} ;
- Y_S is the stress correction factor, which accounts for the amplification of local tooth root stress due to geometric discontinuities and complex stress distributions at the critical section;
- Y_β is the helix angle factor and is equal to 1 for spur gears;
- Y_B is the rim thickness factor, equal to 1 for the gears analyzed in this study, which are not thin-rimmed;
- Y_{DT} is the deep tooth factor, which is equal to 1 for the gears analyzed in this study. Besides, when STB tests like those performed in this study are concerned, the contact ratio is always 1.

The maximum tooth root stress (σ_F) is calculated using the equation (Eq. 2):

$$\sigma_F = \sigma_{F0} \cdot K_A \cdot K_V \cdot K_{F\beta} \cdot K_{F\alpha} \quad (2)$$

where K_A is the application factor, K_V is the dynamic factor, $K_{F\beta}$ is the face load factor for tooth root stress, and $K_{F\alpha}$ is the transverse load factor for tooth root stress. All these factors are equal to 1 in the case of STB Testing.

However, since this testing builds on previous testing [20] the same stress analysis will be employed for stress reporting in this work, and even for those employed in the monotonic testing to define the stress-strain diagram. The

tooth bending stress was analyzed using the Gear Load Distribution and Stress Analysis Program Windows LDP [23] developed by The Ohio State University Gear and Power Transmission Research Laboratory. In this model a mesh position corresponding to the radius of contact determined in the unloaded contact analysis was modeled and a range of mesh force corresponding to the loads used in the experimental testing was input to determine the tooth load distribution. Windows LDP also performs a root stress analysis via Finite Element Analysis (FEA) to determine maximum bending stress [24]. Measured root geometry was imported into the analysis to ensure accurate modeling of the test specimens. The model was based on a linear elastic stress-strain relationship to describe the material behavior. Fig-

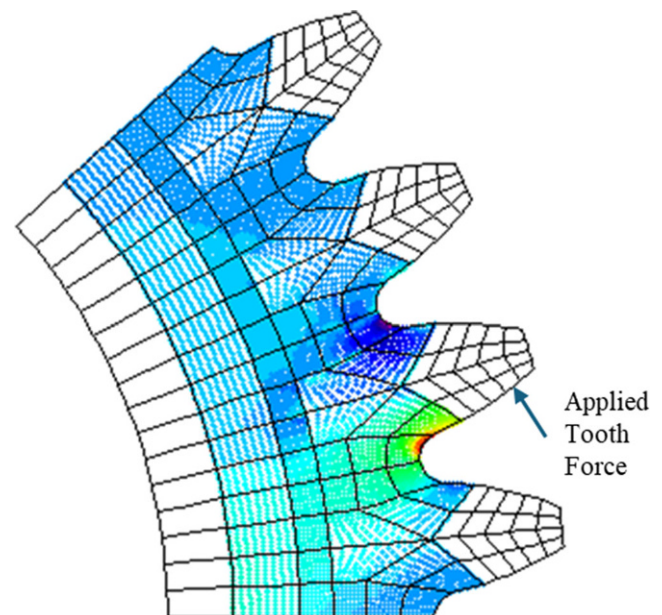


Fig. 5 2D cross section at the center of the face width from the Windows LDP FEA output. Maximum normal (first principal) stress shown as a color map

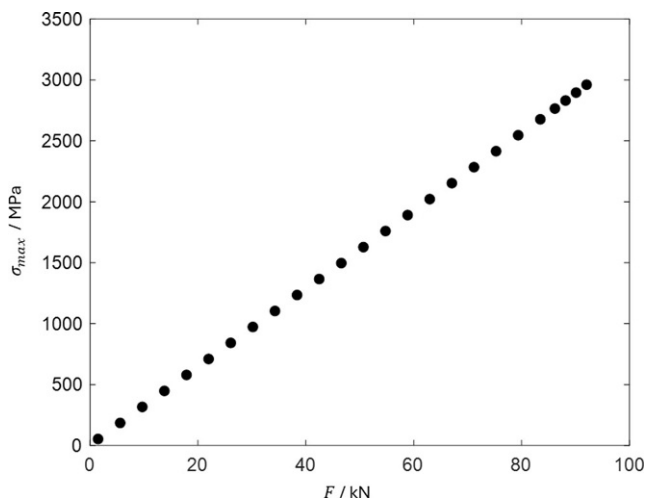


Fig. 6 Maximum normal root bending stress vs applied tooth force predicted from Windows LDP

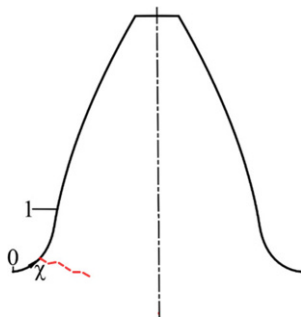


Fig. 7 Critical location analysis strategy

Figure 5 shows a color map of maximum normal stress from a 2D cross section at the center of the face width of four consecutive teeth. Stresses are not shown near the contact zone as the model is not set up to correctly predict those, rather it focuses on tooth root areas.

It could be argued that to properly represent the LCF behavior of the gear a elastic-plastic relationship should be employed, and this is unquestionably true when deformations are concerned. However, the linear elastic estima-

tion used in the FEA was intended exclusively to calculate the maximum bending stress. While deformations obtained for a given load strongly depend on the material behavior, stresses are strictly related to the ratio between load and resistant section, hence depending much less on the type of material response. Moreover, experiments were force controlled, ensuring that the experiments are consistent with the total force applied to the tooth used in the stress prediction model. As a result, the linear elastic estimation used in the FEA is expected only to underpredict the total deformation of the tooth as it does not account for the high strain to stress increment as material in the root begins to strain harden (plasticity). This will produce some difference between predicted and expected maximum stress in the root during experiments, which, however, represents a minor effect on the value of maximum bending stress that is obtained. In addition, using a proportional root stress to applied force assumption remains consistent with the root stress calculation described in the ISO 6336 standard helping to generate a more equivalent and meaningful comparison of the experiment and calculation standard.

As a result, by considering linear elastic deformation, bending stress was found to be satisfactory linear with tooth force, following a proportionality constant between STB tooth force and maximum tooth bending stress equal to 34.35 MPa/kN. Figure 6 shows the relationship of maximum normal root bending stress vs applied tooth force predicted from Windows LDP.

2.4 Critical location analysis

To describe the location in the tooth root where cracks due to TBF occurred (critical location), a methodology employed by the authors in previous works was employed [25, 26]. The critical location on the tooth root fillet is identified through χ , a linear coordinate along the tooth root fillet (Fig. 7). χ is calculated as the length from the beginning of the tooth root fillet (root trochoid center) to the point where failure occurs divided the total tooth root fillet length. When

Fig. 8 Five contour scan measurements (line A to E) are performed along the gear teeth width

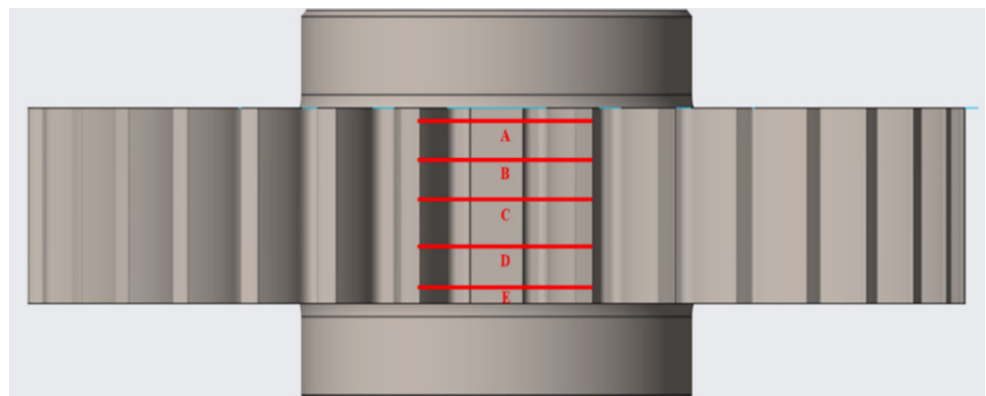


Table 3 Monotonic STB tests results.

Test ID	UTS/kN	E/GPa	σ_Y /MPa
STATIC1	100.44	204.3	2378
STATIC2	90.68	—	—

χ is equal to 0, the critical location is placed at the beginning of the tooth root, while when χ is equal to 1, the critical location is placed at the highest point of the tooth root fillet. If gears are designed in such a way that fillets do not converge at a single point (non-trochoidal root fillets), instead of the root trochoid center it is possible to consider the point on the root circle defining symmetry between two adjacent root fillets and use that as reference point where χ is 0.

To identify the critical location achieved in experimental tests, post-mortem contour scan analysis was performed. A Klingelnberg P26 gear Coordinate Measurement Machine (CMM) was employed to reconstruct the profile of broken teeth. Because TBF fractures have an irregular shape, even if the same tooth is taken into account, the location of the critical point on fillet radius changes along the tooth width. For this reason, for each tooth, five measurements at different points of the tooth width were acquired, as portrayed in Fig. 8 (lines from A to E indicate the five lines along which contour scanning was performed). As a result, a total of ninety contour scan measurements were performed. Both the profile of the fracture surface of the broken tooth and part of the profiles of the adjacent teeth were acquired. χ was calculated for each of the five contour scans that were acquired on each broken tooth.

3 Results

3.1 Monotonic and LCF testing

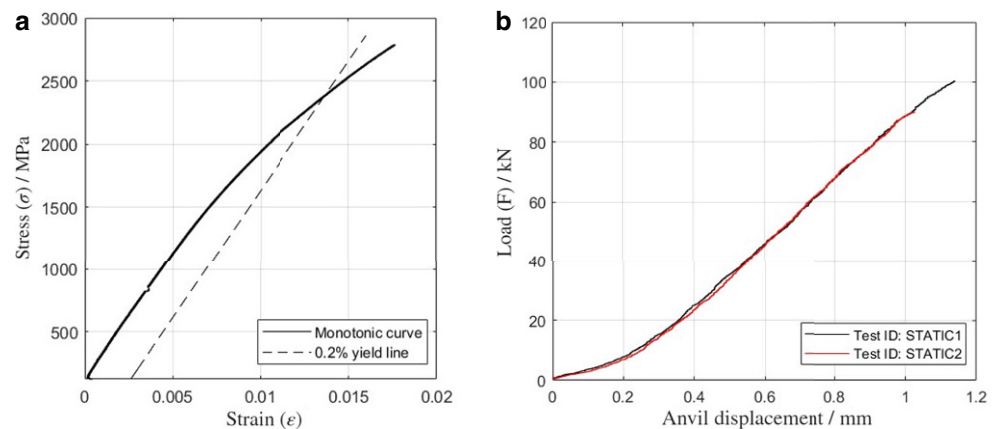
Results of the two monotonic tests reveal a UTS of 90.68 kN for one tooth and 100.44 kN for the other, with an average UTS of 95.65 kN (Table 3). Figure 9a represents the stress-strain diagram that was obtained for the strain-gaged tooth (corresponding to Test ID “STATIC1”). The elastic modulus (E) is 204.3 GPa, while the yield strength (σ_Y) is 2378 MPa. Stresses were calculated according to the proportionality constant equal to 34.35 MPa/kN defined above. Unfortunately, the strain gage did not endure the high plasticity involved in the test and detached before the end of it. The whole load-displacement (of the anvil applying the load) diagrams of the tests are reported in Fig. 9b. Except for the UTS, the two teeth displayed very similar behavior throughout the test.

Results of the LCF STB testing campaign are presented in Table 4 and graphically portrayed in Fig. 13. Tested teeth featured lives spanning a life (N) range between 10^2 and 10^4 cycles, with applied loads ranging from 54 to 84 kN. Stresses were calculated according to the proportionality constant equal to 34.35 MPa/kN defined above. Since the data produced in this study builds upon existing HCF STB data for the same test specimen and material, for continuity of data and direct comparison to the ISO 6336-3 standard (which is based on stresses) a stress-life relationship must be considered, and was therefore preferred to strain-life analysis.

Table 4 LCF STB tests results.

Test ID	Load/kN	σ_F /MPa	Life cycles N	Average χ
LCFG1	84	2885	356	0.51
LCFG2	82	2817	1022	0.59
LCFG3	80	2748	1235	0.43
LCFG4	80	2748	766	0.55
LCFG5	76	2611	1475	0.56
LCFG6	76	2611	1937	0.49
LCFG7	80	2748	2391	0.51
LCFG8	70	2405	2836	0.63
LCFG9	72	2473	1790	0.50
LCFG10	66	2267	3616	0.56
LCFG11	62	2130	3860	0.54
LCFG12	76	2611	1256	0.46
LCFG13	70	2405	1337	0.53
LCFG14	58	1992	9623	0.60
LCFG15	68	2336	2563	0.58
LCFG16	56	1924	11719	0.53
LCFG17	54	1855	8320	0.59
LCFG18	60	2061	5893	0.53

Fig. 9 Stress-strain diagram (a) and load-displacement diagram (b) of monotonic tests



3.2 Critical location

Critical locations detected through contour scanning and quantified through χ as portrayed in Fig. 7 are represented in Table 4. For each test (i.e., for each broken tooth), the value of χ reported in Table 4 is the average of the five χ values obtained through the five contour scan measurements that were performed (line A to E in Fig. 8). Figure 10 represents one example of two of the five contour scans obtained for one of the gear teeth. As can be observed, even if the sharp irregularities of the fracture surface led to some measurement errors in its middle part (top side of Fig. 10), the CMM was able to effectively detect the point on the root fillet where the failure originated. In Fig. 10 it can also be noticed the difference in critical location χ that

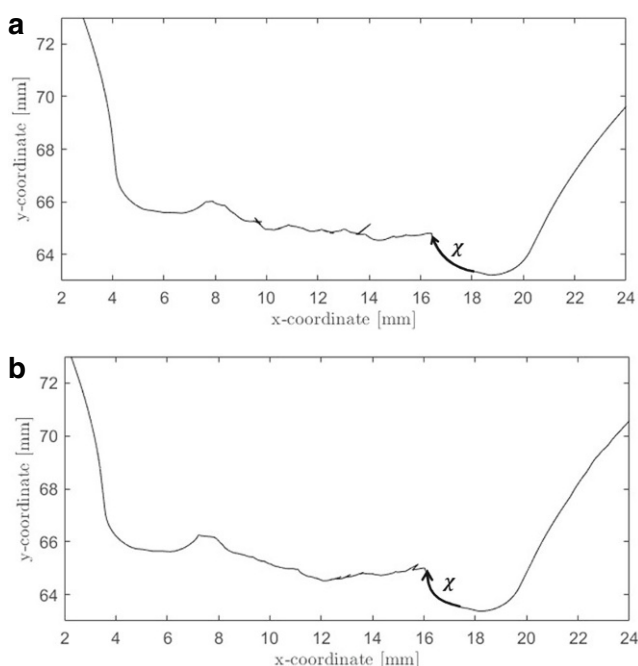


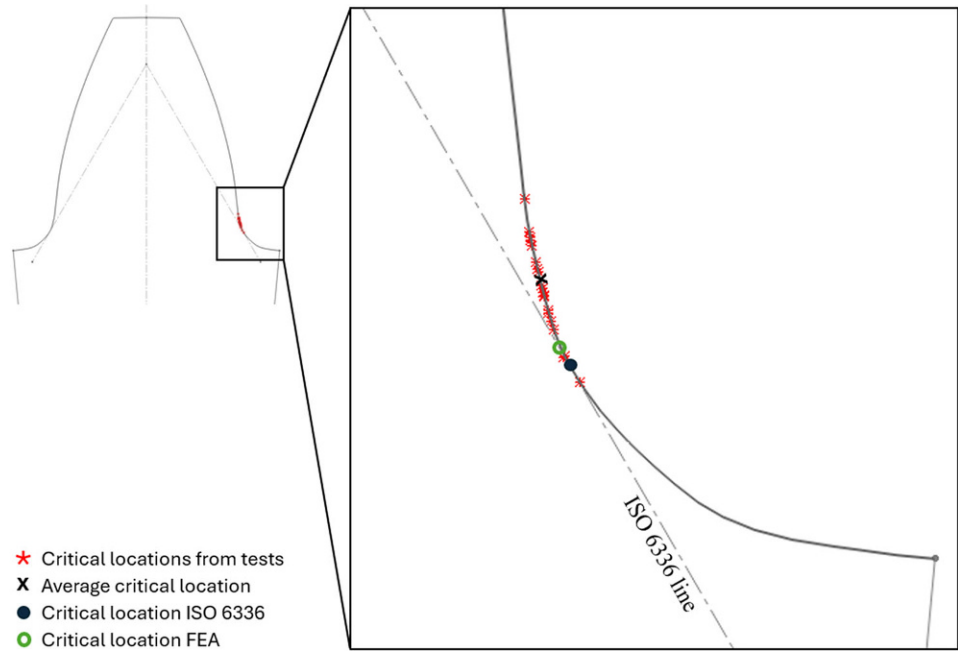
Fig. 10 Contour scan measurement examples from Test ID LCFG17

can be achieved for the same tooth at different points of the tooth width, due to the fracture irregular shape. This is why it was chosen to acquire five contour scans for each one of the eighteen broken teeth (ninety measurements in total) and average the results. The average χ value for all ninety tests (χ_m) is 0.538.

It is also interesting to notice that the average of all χ measured along line A (Fig. 8) and of all χ measured along line E (Fig. 8) for all broken teeth are 0.555 and 0.558 respectively, suggesting that the employed test fixture granted proper and uniform loading of teeth. This is also confirmed by the macroscopic fractography analysis that was performed, as described in Sect. 3.4.

Contour scan results of Table 4 were compared with the critical location that would be obtained following ISO 6336-3 recommendations. The standard prescribes that cracks in the tooth root typically initiate the point of tangency between a line inclined at 30 degrees and the tooth root profile. This guideline is based on empirical observations and analytical studies that have identified this location as the point of maximum tensile stress under load, making it the most susceptible to crack initiation. In Fig. 11 and 12, the critical locations experimentally obtained, their average value χ_m , the critical location predicted by ISO 6336-3 and the one predicted by the FEA model (point of maximum bending stress) are compared. As can be observed, the points predicted by ISO 6336-3 standard and FEA agree (they are both based on a linear-elastic material behavior assumption) and fall within the range of points that were experimentally obtained. However, they lie at the edge of such range, with most of critical locations experimentally obtained located slightly higher in the tooth root fillet than what is predicted by ISO 6336-3 and the FEA model. This is well highlighted in Fig. 12, where the critical location predicted by ISO 6336-3 (χ_{ISO}) and by the FEA model (χ_{FEA}) are compared with the Gauss distribution of critical locations experimentally obtained. The ISO-predicted value falls within the 5.56th percentile.

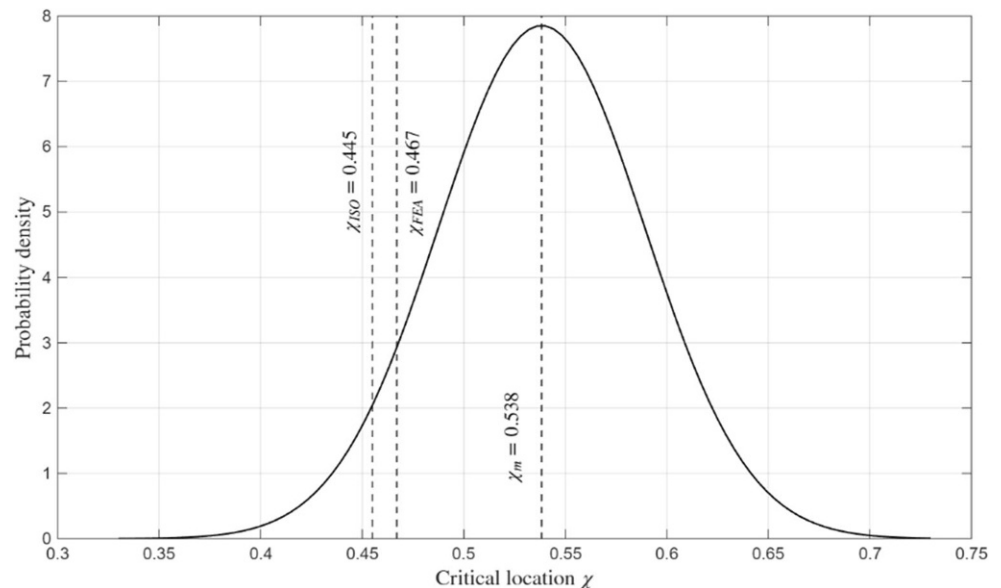
Fig. 11 Comparison of experimental, ISO-predicted and simulation-predicted critical locations.



Even if the difference between actual χ_m and FEA and ISO-predicted critical location might seem an uninfluential detail about teeth behavior, it actually represents a significant aspect for gears fatigue design, influencing the whole crack initiation and propagation trends [27]. The explanation of why such divergency is obtained could lie in the high deformations that the teeth experienced during LCF STB testing. As teeth underwent cyclic loads that exceeded their yield strength, the tooth profile was deformed in such a way that its actual shape during operation deviated considerably from the nominal one, affecting the distribution of stresses and moving the location of maximum bending

stress upwards in the tooth root fillet. Such effect cannot be captured neither by the FEA model, which is based on purely elastic behavior, nor by the method of ISO 6336-3 standard, which is based on the intersection of the 30 degrees-tilted line with the nominal, undeformed profile of the tooth and on the assumption of linear-elastic behavior. Such approaches work effectively in HCF conditions, where deformations are small, but under LCF discrepancies like the one observed in this work might emerge. Also in [28] it is possible to observe a failure location in the higher part of the tooth root fillet due to loads causing plasticity (in this case due to monotonic testing).

Fig. 12 Gaussian distribution of experimental critical locations



3.3 Stress-Life analysis

An examination of the accuracy of the ISO 6336-3 standard for stress-life relationships in the LCF and HCF regime for the AISI 8620 material is performed using the aggregate experimental STB testing (Method A) and tabulated reference values from the standard (Method B). Specifically for comparison, an examination of the LCF and HCF knee points described in the ISO standard Method B is performed where for a case carburized wrought material (material designation: Eh) the LCF knee point occurs at $N = 10^3$ cycles and the HCF knee point occurs at $N = 3 \cdot 10^6$ cycles.

Figure 13 shows the aggregate UTS, LCF, and HCF data [22]. This combined dataset is acceptable since the exact same test equipment, methods and specimens from the same manufacturing batch were used. The new LCF and UTS data are delineated from the previous HCF data to ensure clarity. Visually it is clear that an LCF and HCF knee point exist in this data set, which separates the curve into three distinct regions in agreement with ISO 6336-3 Method B. To generate a quantitative comparison of the ISO standard S–N methods and determine the exact LCF and HCF knee points for Method A, the S–N curve must be fit to the aggregate dataset. For Region 1 the two UTS stresses reported in Table 3 are averaged to determine the placement of the horizontal S–N fit. For Region 2 and 3 the Maximum Likelihood Estimation (MLE) methods described in [22] are utilized. Figure 14 shows the S–N fitting overlayed onto the aggregate STB dataset, with 10%, 50% and 90% failure probability trends. The experimental data

(Method A) results in an LCF knee point that occurs closer to $N = 10^2$ cycles as opposed to $N = 10^3$ described by the ISO standard Method B.

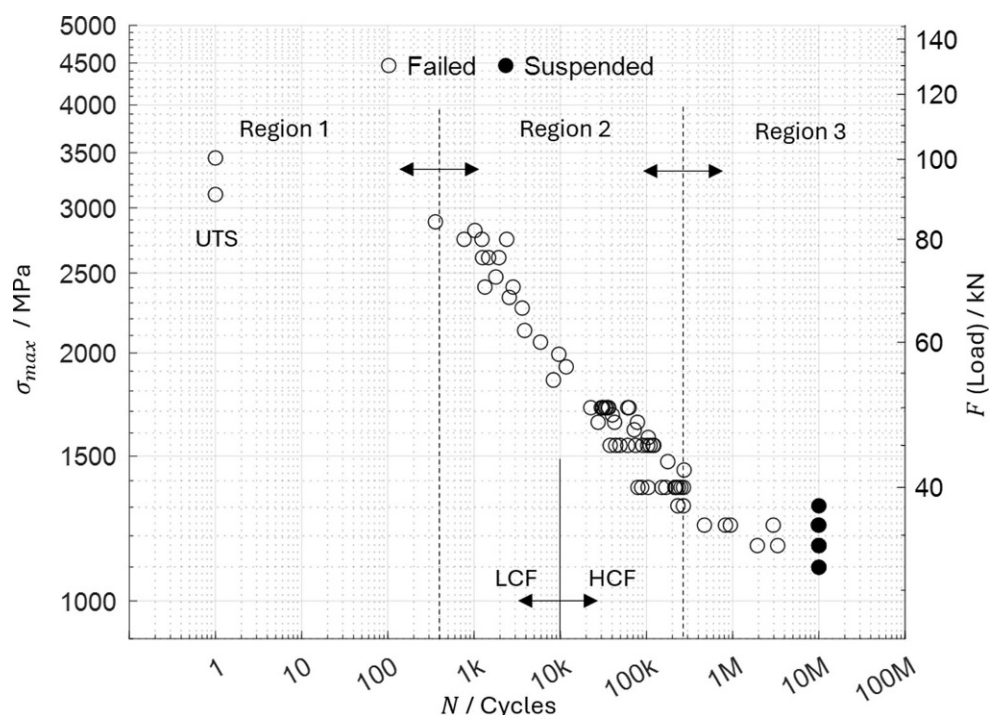
The HCF finite life S–N curve is Basquin type but for practical implementation with reliability takes the form of Eq. 3:

$$\log(\sigma_{\max}) = A + B \log(N) + zS_N \quad (3)$$

Where A and B are the S–N intercept and slope respectively, z is the standard score (z-score) of a gaussian distribution and S_N is the standard deviation defined in cycles and held constant for all valid $\sigma_{\max}$. The log function is defined as the natural logarithm and failure percentiles are calculated by choosing the corresponding z-score to the gaussian cumulative distribution function percentile desired (Table 5).

Two ways to interpret the ISO Method B knee-points are shown. Method B-1 assumes that the LCF and HCF finite life (Region 2) curve is known from experiments and Method B is used for calculating a UTS. Such UTS in Region 1 is drawn as a horizontal line from the MLE finite life at an intersection of $N = 10^3$ for the LCF knee-point. In this case the UTS would be under-predicted as compared to the experimental UTS values. Secondly Method B-2 draws the LCF and HCF finite life curve with the assumption that experimental data exists only for the UTS and HCF Endurance Limit. ISO Method B then instructs that the finite life curve intersects the UTS at $N = 10^3$ for the LCF knee-point and intersects the endurance limit at $N = 3 \cdot 10^6$ cy-

Fig. 13 Combined UTS, LCF and HCF STB data set for case carburized AISI 8620, SAE J1619 test specimens using the dataset in [22]



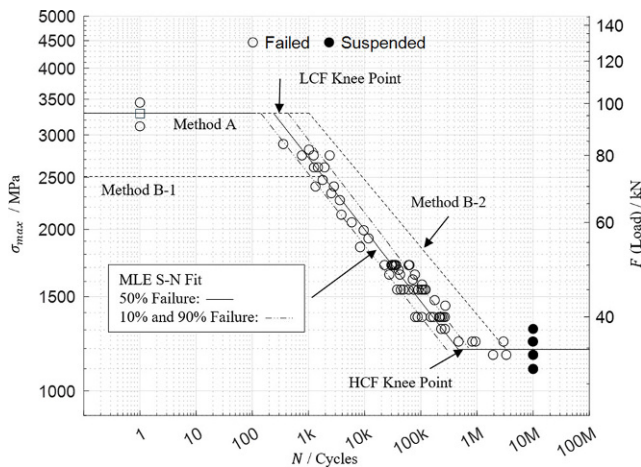


Fig. 14 S–N fitting to the STB dataset

cles for the HCF knee point. In this interpretation the finite life is significantly overpredicted by Method B as compared to the experimental data, but the slope of the resulting finite life is very close to the experimentally derived Method A slope.

The experimental data indicates that the LCF knee point occurs closer to $N = 10^2$ cycles as opposed to $N = 10^3$ described by the ISO standard. In addition, the ISO standard, if rigidly followed, provides a lower, conservative estimate of the allowable stress in the LCF regime for this AISI 8620 case carburized material.

It could be argued that the analysis presented is highly sensitive to the fitted slope of the S–N curve between $10^2 < N < 10^5$. If the slope of the finite life portion is higher, than it is possible to get an LCF knee point that intersects the UTS at a life closer to $N = 10^3$. However, to obtain such a fit requires adjustments to the assumed distribution (Lognormal vs Weibull vs Extreme Value, etc.). Even the most generous fitting to this dataset to increase finite life slope has still resulted in a knee point at a cyclic life lower

Table 5 S–N Fitting Parameters

A	B	S_N	σ_E/MPa
8.8084	−0.13062	0.4143	1196

then $N = 10^3$. Such a fit to get exactly an LCF knee point is shown in Fig. 14. It is clear that it overpredicts the life in the region of $10^2 < N < 10^4$.

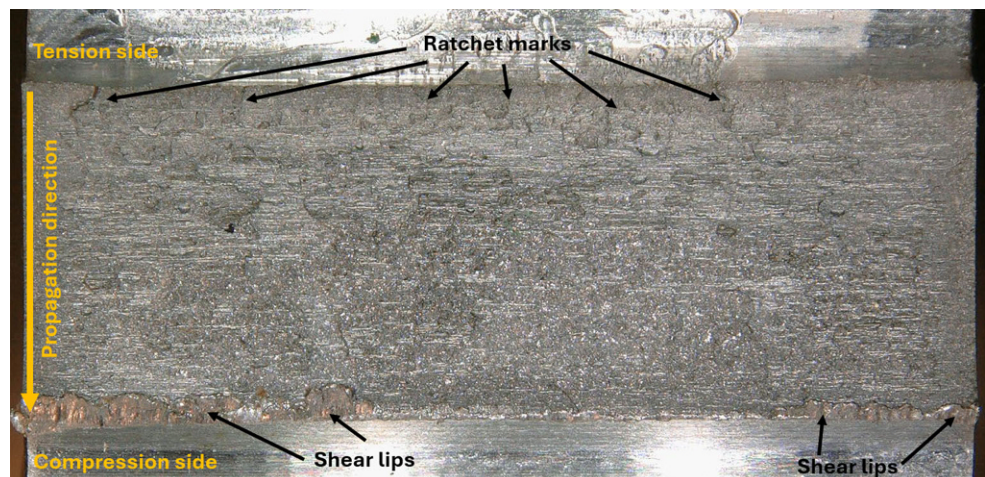
3.4 Fracture surfaces of broken teeth

Macroscopic fractography was performed to document the fracture surface of the LCF fatigue tests. Figure 15 shows a representative fracture surface from test LCFG4. The fracture surface shows a relatively symmetric fracture pattern with respect to the center of the face width indicating a failure origin near the center of the face width. Typical fatigue fracture characteristics are evident such as ratchet marks indicating the intersection of fatigue cracks on different planes [29, 30]. The fracture pattern does not indicate failure occurring from tooth corners (edge breaks) but the wide distribution of ratchet marks centered in the face width indicates a relatively uniform and evenly distribution load applied to the tooth from the fixture. The bottom edge of the image shows some shear lips and rougher fracture surface which are indicative of final overload fracture following fatigue crack growth.

4 Summary and conclusions

An experimental study on gear tooth bending low cycle fatigue has been performed in which eighteen LCF and tests spanning the two decades between $10^2 < N < 10^4$ and two monotonic UTS were completed on SAE J1619 STB specimens made of case carburized AISI 8620 steel. The testing extends a previous sets of experiments performed

Fig. 15 Macroscopic fractography of LCFG 4



in [22], which covered the HCF range of testing using the same specimens from the same manufacturing batch. The purpose of this study is to examine the accuracy of the ISO 6336-3 standard covering bending fatigue load carrying capacity of gears with respect to the prediction of the critical location in the root of the gear as well as the stress-life curve.

To assess the accuracy of ISO 6336-3 in terms of predicted critical location when LCF is involved, contour scan measurements were performed on broken specimens by means of a CMM. The comparison between experimental and ISO Standard-predicted results shows that most of experimentally obtained critical locations lie in a position of the tooth root fillet that is higher than the predicted one. The reason for such discrepancy might lie in the fact that ISO 6336-3 identifies the critical location as the intersection of a line inclined at 30 degrees and the tooth root (undeformed) profile. Because under LCF conditions large deformations take place, the tooth profile during operations deviates significantly from the nominal one, affecting the distribution of stresses and moving the location of maximum bending stress upwards in the tooth root fillet. A very similar result was achieved when comparing experimental results with the critical location predicted by the purely elastic FEA model.

An experimental Stress-Life curve was fitted to the experimental data considering the entire UTS, LCF and HCF data set using Maximum Likelihood Estimates and a log linear fit for the finite life LCF and HCF range. This represents the S–N curve according to ISO Method A. The experimental data shows that the log linear fit accurately predicts the LCF testing results into the range of $10^2 < N < 10^4$. Two routes of incorporating ISO Method B were explored to either (1) predict the UTS strength or (2) predict the finite life behavior. Using the ISO Method B knee-point of $N = 10^3$ underpredicted the UTS and over predicted the finite life fatigue lives.

In conclusion, this work highlights the need for specific design methods and prediction models for gears under LCF. Further experiments should be carried out to enlarge the set of experimental LCF STB data. Moreover, further analysis methodologies (e.g. Finite Element Modelling of gears under LCF loading conditions) should be implemented to evaluate additional aspects, such as critical location estimated through calculated stresses or fatigue life estimated through analytical life prediction models.

Eventually, future work should incorporate strain measurement methods, obtained through techniques like strain gauging or Digital Image Correlation. The inclusion of such measurements is important for validating stress analysis and improving fatigue life predictions. Moreover, evaluating strains is essential for a more comprehensive understanding of local deformation behaviour during LCF STB

tests, and to provide more detailed insight into strain distribution and its evolution in gear components.

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Conflict of interest L. Pagliari, I. Hong and F. Concli declare that they have no competing interests.

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